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Elhanan Buchsbaum

Quantitative Imaging · Image Analysis & Computer Vision · Research Instrumentation

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Research Profile

Experimental scientist (PhD) who builds imaging systems end to end — hardware and acquisition through to the analysis pipelines that turn images into numbers. Most recently built an automated imaging setup and an OpenCV/FFmpeg pipeline with a YOLO detection model to replace manual live/dead scoring of nematodes on 6 cm plates. Earlier, designed and quantitatively validated a six-camera 3D measurement platform (0.5 px reprojection error) with a real-time acquisition layer in Rust, and reproducible analysis pipelines over hundreds of multi-gigabyte recordings on HPC. Comfortable at the bench as well as at the keyboard — immunohistochemistry, fluorescence microscopy and Fiji quantification, plus earlier genome-wide methylation analysis in R/Bioconductor. Two first-author Current Biology papers.

Experience

Postdoctoral Researcher, Max Planck Institute for Neurobiology of Behavior (caesar) – Bonn, Germany — 2025 – 2026

- Built an automated imaging setup for live/dead scoring of nematodes (Lightfoot lab) — camera acquisition, lighting control and a custom mechanical mount — to replace slow manual assessment of 6 cm plates.
- Wrote the acquisition and analysis software — live and offline capture scripts, an OpenCV/FFmpeg processing pipeline, blob extraction, and a YOLO detection and classification model on 5064 × 5064 images. Hardware, acquisition and detection ran end to end within a three-month contract; the classification model reached notebook-stage validation.
- Extended the free-flight imaging platform (Schnell lab) for a forthcoming publication — added optogenetic silencing, improved high-speed acquisition throughput, characterised additional genotypes.

Doctoral Researcher, MPI for Neurobiology of Behavior (caesar) / University of Bonn – Bonn, Germany — 2019 – 2025

- Designed and built a closed-loop measurement platform end to end — multi-camera high-speed imaging, real-time 3D tracking at 100 Hz, optogenetic light control, visual displays and microcontroller triggering, synchronised at 10 ms closed-loop latency.
- Built and validated the multi-camera 3D measurement stack — intrinsic and extrinsic calibration across six cameras at 100 fps, triangulation, and calibration validated quantitatively at 0.5 px reprojection error against a defined accuracy target.
- Wrote the timing-critical acquisition and triggering layer in Rust after a Python implementation could not meet latency requirements; Python for all surrounding tooling and analysis.
- Developed reproducible Python pipelines processing hundreds of recordings of ∼10 GB each in HDF5/Parquet, parallelised locally and run as Slurm batch jobs.
- Applied markerless pose estimation (DeepLabCut, SLEAP) to high-speed, high-resolution video, then segmented and classified trajectories using clustering and mixture models.
- Verified GAL4 driver-line expression and confirmed the identity of individual neurons by immunohistochemistry on dissected brains — primary/secondary and nanobody-based antibody staining, fluorescence microscopy, and quantification in ImageJ/Fiji across several hundred animals.
- Documented the rig and shipped self-service analysis tools so lab members without programming backgrounds could run standardised experiments and analyses unsupervised; trained and supported them, and supervised one MSc student.
- **Result:** first-author paper in Current Biology (2025) identifying a descending neuron that drives visually evoked flight saccades. The platform remains in daily use as one of the group's standard experimental tools.

Research Rotations, caesar – Bonn, Germany — 2018 – 2019

- Two-photon calcium imaging in behaving Drosophila (Seelig lab); analysis of two- and three-photon imaging data in rodents (Kerr lab).

MSc Researcher, Technion – Israel Institute of Technology, Faculty of Medicine – Haifa, Israel 2016 – 2018

- Built MATLAB signal-processing and analysis pipelines for in vivo tetrode electrophysiology — spike sorting, single-unit and spatial-tuning analysis, and statistics over large multi-dimensional time-series (~2,100 sorted units across 23 animals).
- Modernised and extended the lab's legacy analysis codebase, documented the procedures and trained incoming researchers; the tooling was adopted lab-wide.
- Performed the experimental work hands-on — microsurgery, tetrode and microdrive construction, stereotactic implantation — adapting recording hardware designed for rodents to an avian species. Co-first-author paper in Current Biology (2021).

Research Assistant — Bioinformatics and Behavioural Neuroscience, University of Haifa – Haifa, Israel 2014 – 2016

- Genome-wide differential methylation analysis in R/Bioconductor — FDR-corrected workflows to identify differentially methylated regions and candidate genes, pathway enrichment (Ingenuity Pathway Analysis) and qPCR validation. Wet-lab molecular work — DNA/RNA extraction, qPCR, cell culture, gene-expression analysis.
- Histological characterisation of a rat model of PTSD (Richter-Levin lab) — cryosectioning, immunohistochemical staining, microscopy and manual quantification of labelled cells. Co-authored paper in Molecular and Cellular Neuroscience (2021).

Selected Projects

OptoFly — closed-loop tracking and optogenetic stimulation platform – public, GPL-3.0 2020 – 2026

- Integrates multi-camera 3D tracking with triggered recording, optogenetic stimulation, dynamic autofocus and configurable visual stimuli; Rust acquisition layer with Python bindings. 485 commits, documented via a live docs site.

Technical Skills

Imaging & microscopy: High-resolution plate and live imaging, multi-camera acquisition, intrinsic/extrinsic camera calibration and validation, machine-vision cameras (Basler/Pylon), lens selection, LED/IR illumination, two-photon calcium imaging, fluorescence microscopy, ImageJ/Fiji

Image analysis & ML: OpenCV, scikit-image, FFmpeg, object detection (YOLO), markerless pose tracking (DeepLabCut, SLEAP), scikit-learn, clustering and mixture models, statistical modelling

Programming: Python (primary), Rust (real-time acquisition), MATLAB, R, Bash, Arduino/C, Git/GitHub

Data & infrastructure: HDF5, Parquet, Linux, Docker, HPC (Slurm), reproducible parallel/batch pipelines, large-scale image and video datasets

Instrumentation: Closed-loop real-time control, hardware triggering and synchronisation, microcontrollers, custom rig design, experimental design with defined acceptance criteria

Wet lab & molecular: Immunohistochemistry (primary/secondary and nanobody staining), cryosectioning, microsurgery, DNA/RNA extraction, qPCR, cell culture, R/Bioconductor differential methylation and pathway enrichment

AI-assisted development: LLM-assisted development (Claude Code), MCP-based tool integration, agentic coding harnesses, OpenAI-compatible APIs

Education

University of Bonn / MPI for Neurobiology of Behavior (caesar), PhD in Neuroscience – Bonn, Germany 2019 – 2025

- magna cum laude. IMPRS for Brain and Behavior.
- Thesis: *Neural Control of Flight Maneuvers in Freely Flying Drosophila melanogaster* (advisor: Dr. Bettina Schnell).

Technion – Israel Institute of Technology, MSc in Neuroscience – Haifa, Israel 2016 – 2018

- summa cum laude.
- Thesis: *Space Coding in the Japanese Quail Hippocampal Formation* (advisor: Prof. Yoram Gutfreund).

University of Haifa, BSc in Biology and Psychology – Haifa, Israel 2013 – 2016

- Bioinformatics research focus. Thesis: *Epigenetic Modulation of Autism Occurrence in 15q11.2 Deletion Carriers* (advisor: Prof. Gil Atzmon).

Publications

Activity of a descending neuron associated with visually elicited flight saccades in Drosophila 2025
Elhanan Buchsbaum, Bettina Schnell
10.1016/j.cub.2024.12.001 (Current Biology 35(3), 665–671)

Directional tuning in the hippocampal formation of birds 2021
Co-first author (with K. Krivoruchko).
Elhanan Ben-Yishay, K. Krivoruchko, S. Ron, N. Ulanovsky, D. Derdikman, Y. Gutfreund
10.1016/j.cub.2021.04.029 (Current Biology 31(12), 2592–2602)

Behavioral profiling reveals an enhancement of dentate gyrus paired pulse inhibition in a rat model of PTSD 2021
Work published before 2022 appears under the former surname Ben-Yishay.
A. Albrecht, **Elhanan Ben-Yishay**, G. Richter-Levin
10.1016/j.mcn.2021.103601 (Molecular and Cellular Neuroscience 111, 103601)

Fellowships & Training

- Max Planck Minerva Fellowship (2021); IMPRS Fellowship, doctoral selection (2019).
- Janelia Mechanistic Cognition Junior Scientist Workshop (2025); FENS-Hertie Winter School, Neural Control of Behavior (2017).

Referees

- Dr. Bettina Schnell — Max Planck Institute for Neurobiology of Behavior (caesar), Bonn. Doctoral advisor. [add email]
- Dr. James Lightfoot — Max Planck Institute for Neurobiology of Behavior (caesar), Bonn. Host for the nematode imaging project. [add email]
- Prof. Yoram Gutfreund — Technion – Israel Institute of Technology, Haifa. MSc advisor. [add email]